

Bahaa Hamed

Bachelor of Engineering – Computer Science & Economics | AI / Backend / Systems Development

bahaahamed1970@gmail.com | +46 720 194 389 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFILE

I am a newly graduated Bachelor of Engineering in Computer Science and Economics from KTH (2026) with a strong interest in AI and how it can solve real-world problems. I have a passion for technology, innovation, and social impact. With solid skills in programming, databases, and systems development as well as hands-on project experience, I have a strong foundation to build on. During my thesis project at Epinova AB I built Archivon – a multi-agent AI system from scratch – and that is where I truly discovered how exciting it is to work at the intersection of AI, backend development, and system design. I enjoy taking ownership, learning quickly, and contributing to teams that build things that actually get used. I am now looking for a role where I can continue to grow within software development and AI.

EDUCATION

Bachelor of Engineering – Computer Science & Economics | KTH Royal Institute of Technology | 2022–2026

- Bachelor's degree with thesis project (15 credits) focused on AI-driven documentation automation.
- Relevant courses: Algorithms & Data Structures, Databases, Machine Learning, Software Engineering, Economics & Business Development.

THESIS PROJECT – ARCHIVON (2026)

Archivon – Distributed RAG-Based Multi-Agent AI System | KTH × Epinova

- Designed and implemented a complete AI system for automatic documentation generation, adapted per target audience (persona-driven output).
- Ingests artifacts from GitHub (PRs, commits) and Azure DevOps (user stories) via custom-built connectors.
- RAG pipeline: text-embedding (Azure OpenAI), pgvector (PostgreSQL), cosine similarity retrieval with automatic grouping strategy.
- Architecture: Two .NET 10 microservices (archivon-core, archivon-generator), exposed via MCP SSE streaming and integrated with the Optimizely Opal Tools SDK.
- Deployed on Azure App Service (Sweden Central) with CI/CD via GitHub Actions (path-filter workflows per agent).
- Prompt injection defence via sandwich pattern; project name resolution with Levenshtein distance fallback.

Tech stack: C# .NET 10, PostgreSQL, pgvector, Azure OpenAI, Azure App Service, GitHub Actions, MCP/SSE, RAG, Optimizely Opal SDK

OTHER PROJECTS

Admin Panel for Wait App – Patient Management System | KI / KTH | 2024

- Built an admin panel for managing and visualising patient data in real time, developed in collaboration with Karolinska Institutet.
- Built the full-stack solution using React (TypeScript) on the frontend and Quarkus (Java) on the backend, connected to PostgreSQL.
- Implemented secure authentication with Keycloak, a REST API for data communication, and dynamic charts for blood pressure monitoring.
- Delivered search and sorting features to streamline staff workflows.
- Tech stack: React (TypeScript), Java, Quarkus, PostgreSQL, Docker, Keycloak, Axios, Jest

Ball Brawl – Networked 2D Multiplayer Game | KTH | 2023

- Built a 2D football game where up to four players compete against each other over a network in real time.
- Responsible for the TCP/IP network communication as well as the game logic and physics.

- Structured the codebase using MVC architecture to separate game logic, graphics, and the network layer.
- Delivered a stable prototype with a responsive gameplay experience and minimal latency.
- Tech stack: C, SDL2, SDL_net, TCP/IP, MVC, GitHub, Scrum

Drive My Kid – Carpooling App (UX) | KTH | 2023

- Designed a mobile app that helps parents organise carpooling for their children to school and activities.
- Conducted needs analysis and interviews with the target group to understand user needs and safety requirements.
- Created interactive Figma prototypes and iterated the design based on usability tests.
- Resulted in an intuitive interface with built-in safety features to keep children's journeys secure.
- Tech stack: Figma, UX Research, Usability Testing, Design Thinking

TECHNICAL SKILLS

AI & ML: RAG pipelines, vector databases (pgvector), embeddings (Azure OpenAI), LLM integration, prompt engineering, multi-agent systems

Backend: C# .NET 10, Java (Quarkus), MCP/SSE streaming, REST API, microservices

Programming: C#, Java, JavaScript, TypeScript, C, Assembly, VHDL

Databases: PostgreSQL, pgvector, MySQL, MongoDB

Cloud & DevOps: Azure (App Service, OpenAI, DevOps, PostgreSQL), GitHub Actions, Docker

Web & Frameworks: React, HTML/CSS, Quarkus, MVC

Tools: Figma, Keycloak, Axios, Jest, Scrum/Agile, ngrok

Soft Skills: Problem solving, communication, teamwork, adaptability

EXPERIENCE

Personal Assistant | Humana AB | 2018 – present

- Provided high-quality support and care to individuals with varying needs.
- Built trust through empathy, attentiveness, and adaptability.

Picker & Packer | Zalando | Jan 2021 – Jun 2022

- Handled customer orders efficiently in a fast-paced logistics environment.

IT Technician | Computer Engineering Company | Jul 2013 – Jan 2015

- Customer support, technical troubleshooting, and retail sales of hardware and software.

LANGUAGES

Swedish: Fluent

English: Fluent

Arabic: Native